

SAI MEDH

Hyderabad, Telangana, India • github.com/saimedh • Portfolio: saimedh.dev • AI / ML & Software Developer

EDUCATION

CMR Institute of Technology, Hyderabad — *B.Tech in Computer Science and Engineering (AI & ML)* | 2024 – Present

- Core Coursework: Machine Learning Foundations, Deep Learning, Data Structures & Algorithms, Database Systems, Cloud Computing.

TECHNICAL SKILLS

AI / Machine Learning: PyTorch, scikit-learn, XGBoost, LLM APIs (OpenAI, Gemini), RAG, Prompt Engineering, Embeddings, GeoPandas

Backend & Systems: Python, FastAPI, Node.js, PostgreSQL, Redis, RESTful API Design, Docker, Kubernetes, Linux

Frontend & UI: React, TypeScript, Next.js, Tailwind CSS, Vite, Streamlit

Tools & Platforms: Git, GitHub Actions, AWS, Supabase, Firebase, Nginx, MCP (Model Context Protocol)

FEATURED PROJECTS

Vedha AI (Paverasa AI) — Unified AI Productivity Workspace & Workflow Engine | *FastAPI, Flutter, Redis, PostgreSQL, Gemini/OpenAI APIs, Docker*

- Architected full-stack AI workspace unifying 10+ AI tools (AI Chat, Code Generator, PDF Chat, OCR Scanner, Image Gen) eliminating tool-switching fatigue.
- Implemented resilient FastAPI microservice layer with Redis caching, asynchronous task queues, upstream rate limiting, and automated multi-model fallbacks.
- Containerized backend services with Docker & Kubernetes for production deployment on AWS behind Nginx reverse proxy with automated GitHub Actions CI/CD.

India Crime Rate Prediction System — Spatiotemporal Patrol Risk Forecasting | *Python, XGBoost, scikit-learn, GeoPandas, Streamlit*

- Processed and cleaned 3,600+ historical incident records across 20 cities and 16 crime categories, engineering 37 spatiotemporal features (rolling density, time-of-day bins).
- Trained and cross-validated XGBoost classification and regression models, achieving ~93% risk classification accuracy and $R^2 \approx 0.92$ incident frequency forecasting.
- Deployed interactive GIS mapping dashboard enabling urban civic safety teams to simulate risk scenarios and optimize patrol unit allocations.

Health Navigator AI — Clinical Symptom Triage & Specialist Routing | *Python, FastAPI, PyTorch, Redis, React, TypeScript*

- Built clinical symptom-to-specialist routing engine fine-tuned on medical taxonomy benchmarks across 200 validation scenarios with 83% agreement.
- Engineered deterministic rule-based emergency triage filters prior to model inference, guaranteeing zero missed red-flag emergency symptoms.
- Created touch-optimized clinic kiosk frontend with Redis session caching and clean clinical intake handoff documentation.

FreeKeys (API Emporium) — Directory & Comparison Tool for Free AI APIs | *React, TypeScript, Tailwind CSS, Vite, Model Context Protocol*

- Developed searchable developer directory indexing 90+ free AI model API tiers and 40+ Model Context Protocol (MCP) servers with real-time rate limit comparisons.

EXPERIENCE

AI / Software Developer (Independent & Client Projects) | 2025 – Present

- Shipped multiple live production AI web applications with end-to-end data pipelines, custom ML models, and API integrations.